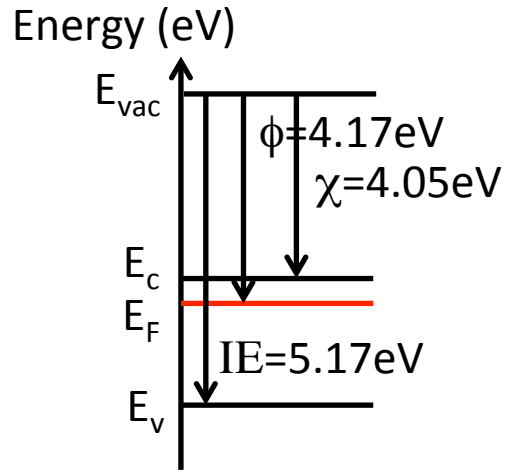
EEEN20070 tutorial questions
Set 2: Junctions – a combination of materials

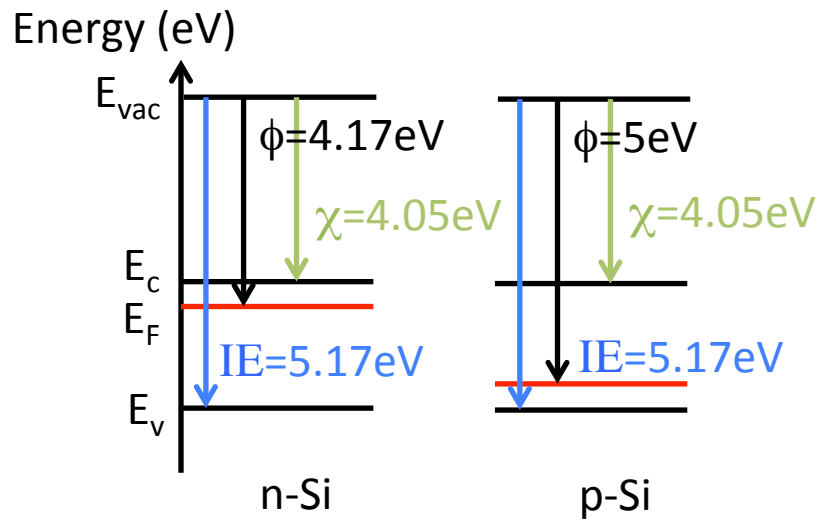
1. First the calculations:
   $n = N_c \exp[(E_F - E_c)/kT]$ from formulae sheet, N_c in constants list (see first SG question sheet)
   $n = N_D$ (complete ionisation of the doping atoms)
   Rewrite equation:
   $E_c - E_F = kT \ln(N_c/N_D)$
   $E_c - E_F = 0.026 \text{ eV } \ln(3.2 \times 10^{19} \text{ cm}^{-3}/3\ 10^{17} \text{ cm}^{-3}) = 0.12 \text{ eV}$
   Workfunction is:
   $\chi + (E_c - E_F) = (4.05 + 0.12) \text{ eV} = 4.17 \text{ eV}$.



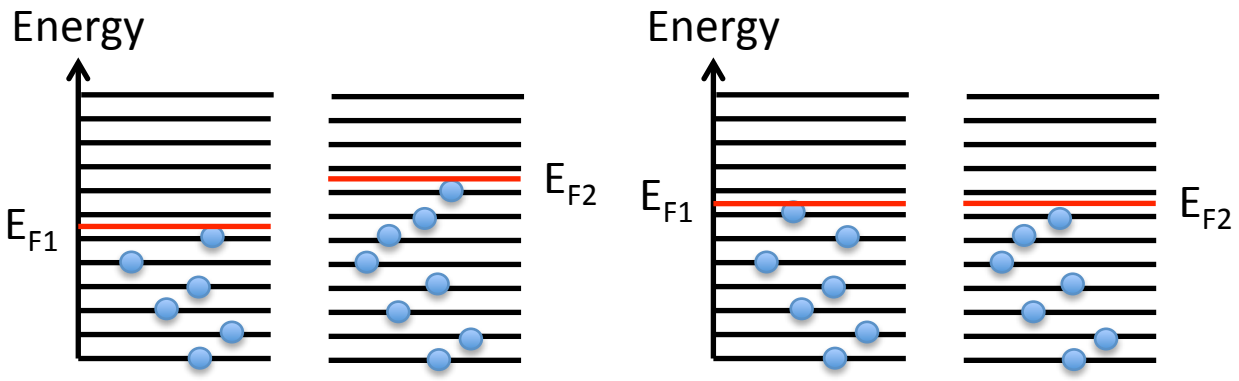
   IE is the ionisation energy (cfr. Chemistry)

2. We know the energy band diagram of the n-type material from question 1. We need to do the calculations for the p-type material.

   $n = N_c \exp[(E_F - E_c)/kT]$
   $p = N_A$ and $n = n_i^2/N_A$
   Rewrite equation:
   $E_c - E_F = kT \ln(N_c \times N_A/n_i^2)$
   $E_c - E_F = 0.026 \text{ eV } \ln(3.2 \times 10^{19} \times 5\ 10^{16}/(1.45\ 10^{10})^2) = 0.95 \text{eV}$
   Workfunction is:
   $\chi + (E_c - E_F) = (4.05 + 0.95) \text{ eV} = 5 \text{eV}$.

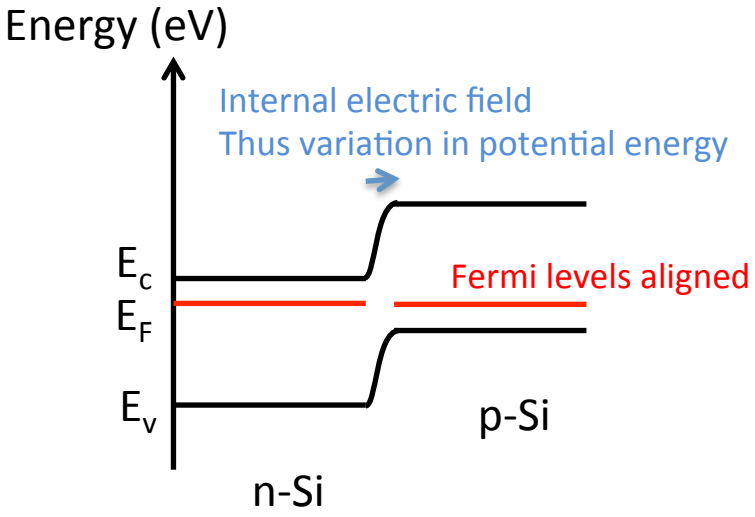


3. Consider two separate materials with filled and empty energy levels determined by the position of the Fermi level, E_{F1} and E_{F2} as given in Fig. 2.1. Re-draw the filled and empty spaces of these two materials when they are brought into contact.



Right figure is the two materials in contact. E_{F1} moved up and E_{F2} moved down. EF levels are now aligned such that no carriers are available at higher levels than the lowest un-occupied level in the combination.

4. Draw the energy band diagram of the materials in question 2 when they are in contact.



5. Sketch the energy band diagram (E_c, E_v, E_F, E_G) of a Si np diode when no external bias is applied. Calculate the position of the conduction band, Fermi level and intrinsic level relative to the valence band and the value of the built-in voltage for a diode with acceptor doping $N_A=10^{17}$ cm^{-3}, donor doping $N_D=10^{17}$ cm^{-3}, and bandgap of bulk Si is $E_G\approx1$eV at room temperature.

Calculation of the different parameters.
in n-region : free electron concentration

$$n_n = N_D$$

$$n_n = N_c \exp\left(-\frac{(E_{c_n} - E_{F_n})}{kT}\right)$$

$$\Rightarrow E_{c_n} - E_{F_n} = kT \ln \frac{N_c}{N_D}$$

$$E_{c_n} - E_{F_n} = 0.026\,eV \ln\left(\frac{3.2\ 10^{19}\,cm^{-3}}{10^{17}cm^{-3}}\right)$$

$$E_{c_n} - E_{F_n} \simeq 0.15\ eV$$

$$n_n = n_i \exp\left[\frac{E_{F_n} - E_{i_n}}{kT}\right]$$ see problem sheet 1

$$\Rightarrow E_{F_n} - E_{i_n} = kT \ln\frac{n_n}{n_i} = kT \ln\left(\frac{10^{17}\,cm^{-3}}{1.45\ 10^{10}cm^{-3}}\right)$$

$$\simeq 0.41\ eV$$

in p-Region : free hole concentration

$$p_p = N_A$$

$$p_p = N_v \exp\left(\frac{-(E_{F_p} - E_{v_p})}{kT}\right)$$

$$E_{F_p} - E_{v_p} = kT \ln\left(\frac{N_v}{N_A}\right) = 0.026 \ln\left(\frac{1.8\ 10^{19}}{10^{17}}\right)$$

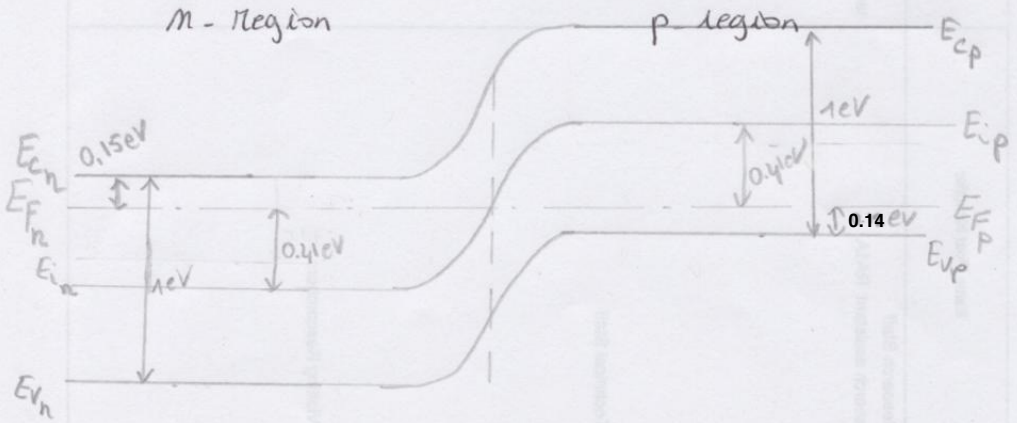
$$E_{F_p} - E_{v_p} \simeq 0.14\ eV$$

$$p_p = n_i \exp\left[\frac{E_{i_p} - E_{F_p}}{kT}\right]$$

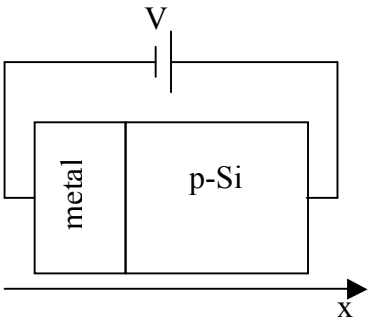
$$E_{i_p} - E_{F_p} = kT \ln\left(\frac{p_p}{n_i}\right) = 0.026 \ln\left(\frac{10^{17}\,cm^{-3}}{1.45\ 10^{10}cm^{-3}}\right)$$

$$\simeq 0.41\ eV$$

Band diagram @ $V_{ext} = 0$ $\Rightarrow$ E_F is constant throughout structure.



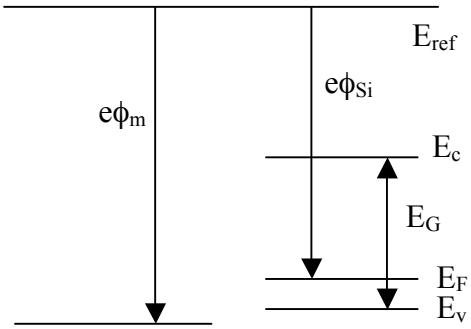
6. *Metal-semiconductor contact*



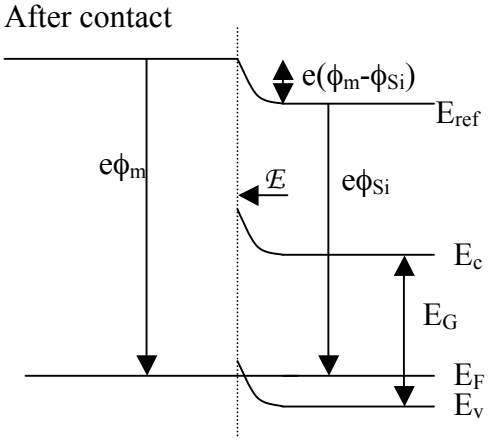
Workfunction relationship
$\phi_m > \phi_{Si}$

a. *Draw the energy band diagram (E_c, E_v, E_F, E_G) through the metal semiconductor junction for V=0V.*

Before contact

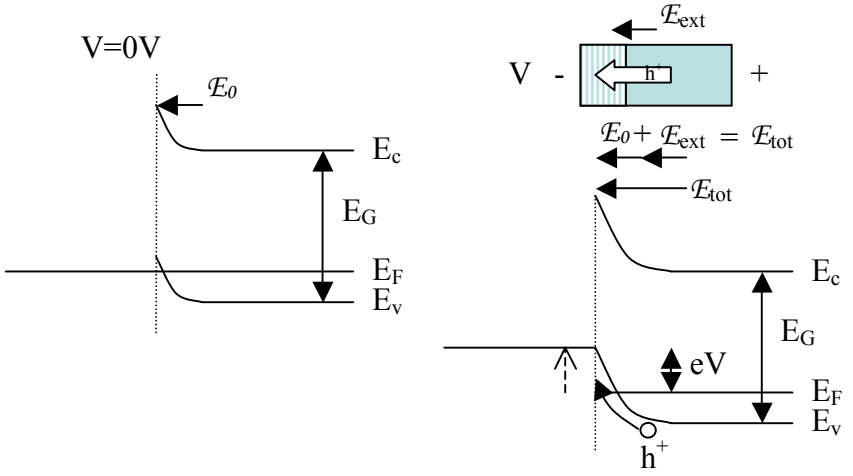


After contact

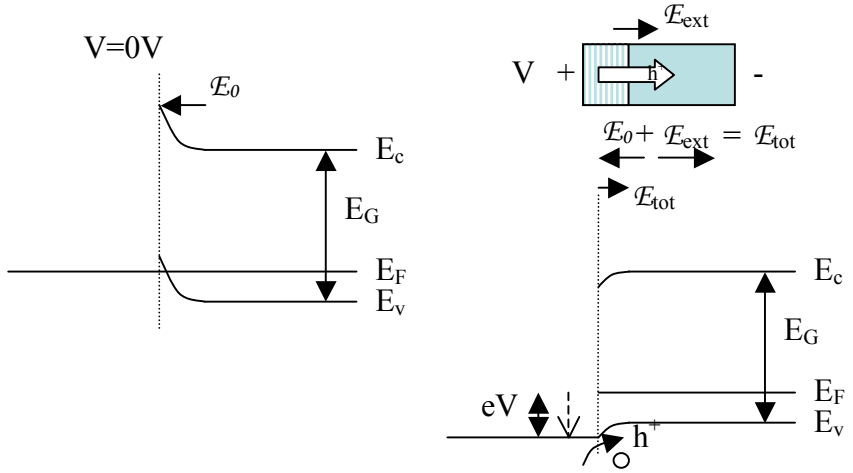


Upon contact diffusion of electrons happen from the semiconductor to the metal. Since V=0V and thus I=0A there needs to exist an electric field $\mathcal{E}$ across the junction to create drift of carriers of the same amount as the carriers diffusing over. The electric field is equal to the contact potential difference: $\phi_m-\phi_{Si}$.
Note that the energy bands are bending due to this internal electric field. This means that the concentration of holes at the junction (dashed line) is larger than the concentration of holes far from the junctions. Thus we have accumulated holes – majority carriers in a p-type semiconductor – at the interface (junction).

b. *Draw the energy band diagram (E_c, E_v, E_F, E_G) through the metal semiconductor junction for V as given in the figure above.*
Note that holes with lower potential energy are closer to the top of the valence band. So when holes lose potential energy they go up toward E_v, thus they flow uphill like air bubbles in water.
Plot the direction of the extrinsic electric field. In this case it is the same as direction as the internal extrinsic field so it will allow more holes to flow towards the junction and into the metal.



c. Draw the energy band diagram (E_c, E_v, E_F, E_G) through the metal semiconductor junction for a polarity of V opposite to that given in the figure above.



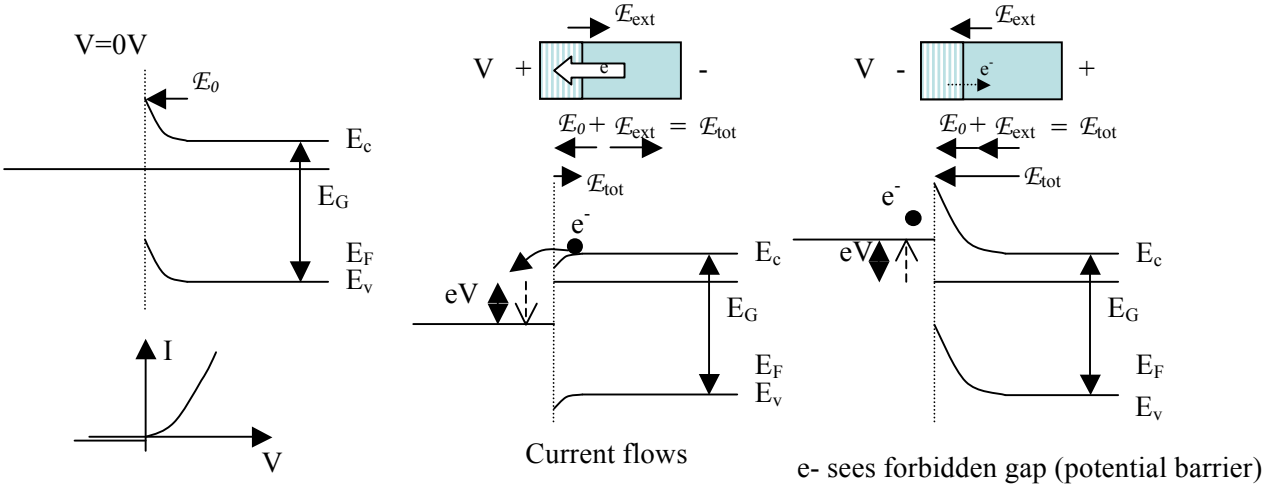
d. What type of contact does this metal-semiconductor junction make?

i. Ohmic contact

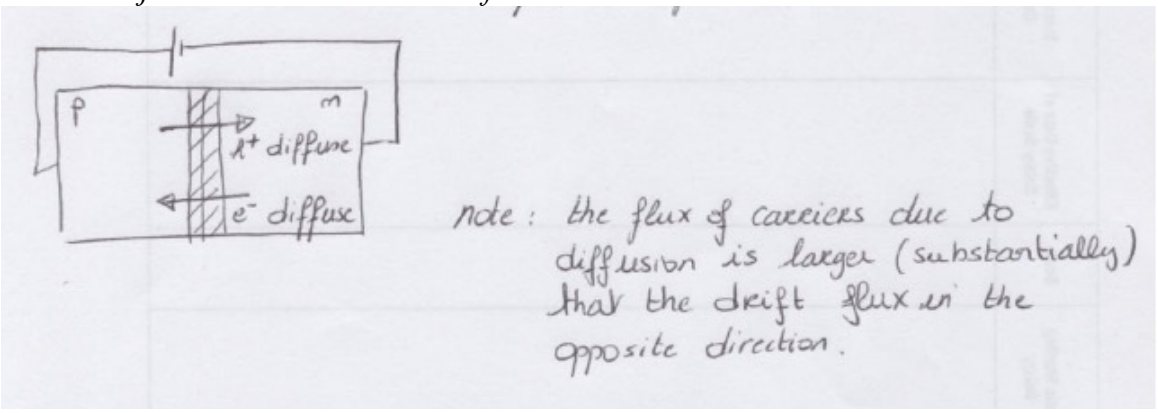
ii. For an ohmic contact the metal is always chosen such that the contact potential increases the majority carrier concentration at the interface.

e. What would happen to the current-voltage characteristic of the contact if we change the silicon to n-type for the same workfunction relationship?

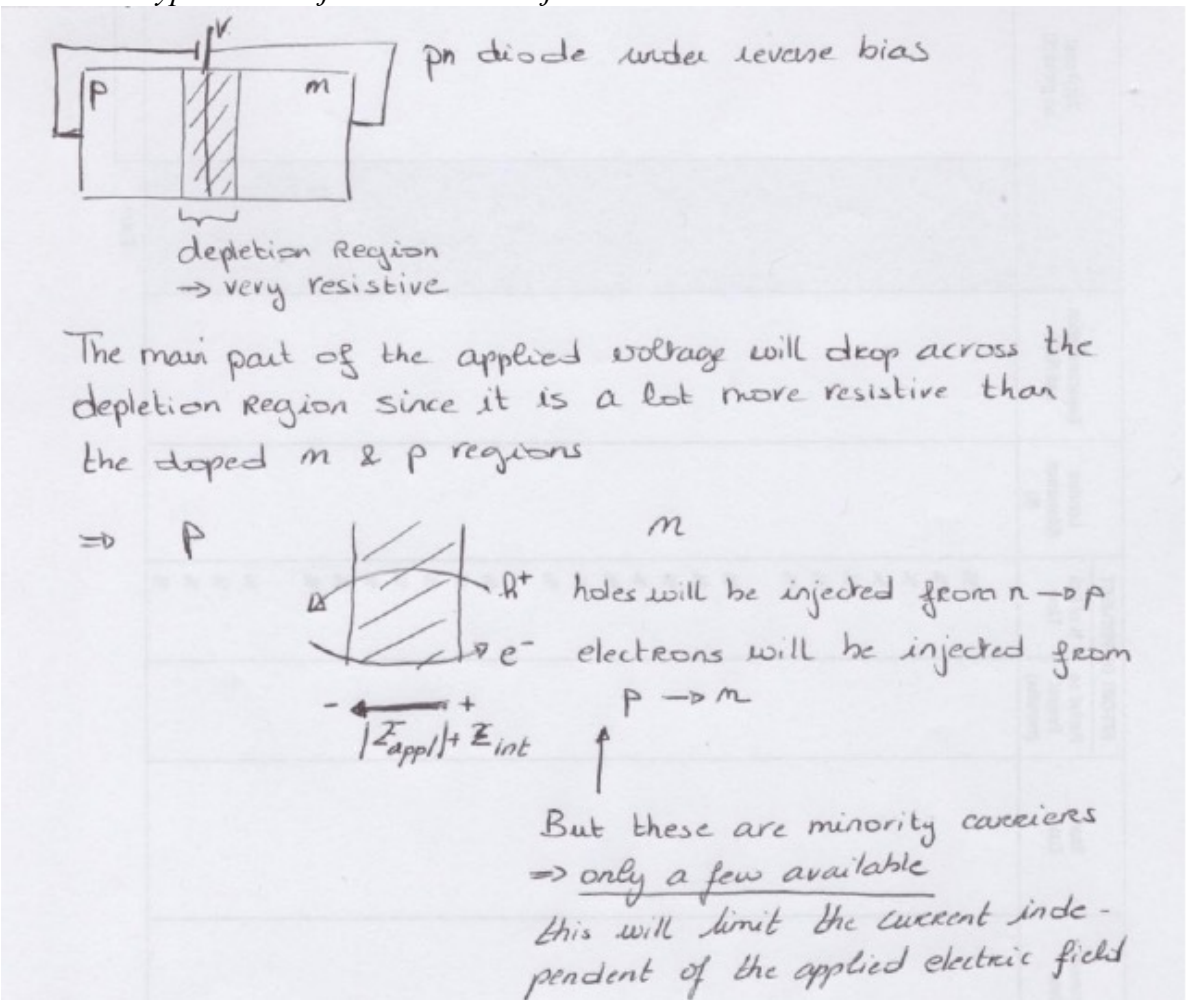
Then a Schottky contact would be formed.



7. *On a schematic diagram of a pn diode, draw the direction of the hole and electron flow across the junction when the diode is forward biased.*



8. *Why is the current in a reverse biased pn diode small? Explain your answer in function of the carrier type that is injected across the junction.*



9. Sketch the variation, if any, of the minority and majority carrier concentration as a function of distance in a short Si pn diode with acceptor doping $N_A=10^{15}$ cm^{-3}, donor doping $N_D=10^{18}$ cm^{-3}, and bandgap $E_G\approx1$eV at room temperature when:
    a. The diode is unbiased
    b. The diode is under a forward bias of 0.5V
    c. The diode is under a reverse bias of 1V

(3)

p-Region: majority carriers $p_p = N_A = 10^{15}\,cm^{-3}$

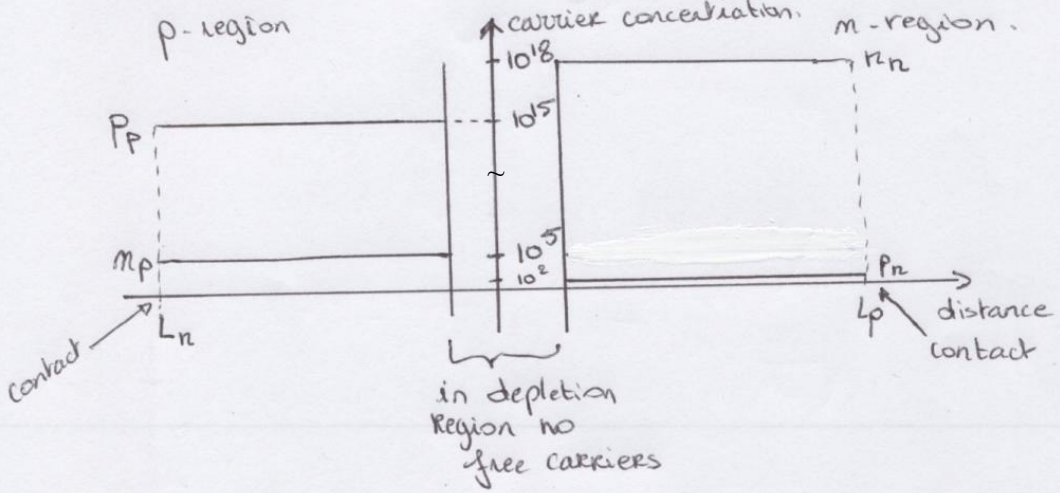
minority carriers $n_p = \frac{n_i^2}{N_A} = \frac{(1.45\,10^{10})^2}{10^{15}} = 2.1\,10^5\,cm^{-3}$

n-Region: majority carriers $n_n = N_D = 10^{18}\,cm^{-3}$

minority carriers $p_n = \frac{n_i^2}{N_D} = \frac{(1.45\,10^{10})^2}{10^{18}} = 210\,cm^{-3}$

a) unbiased: $V_{ext} = 0$

⇒ no excess minority carriers.



b) forward biased: $V_{ext} = 0.5V$

see equation sheet: $c = c_0 \exp\left(\frac{eV}{KT}\right)$

→ excess concentration will be generated via the injection of majority carrier across the junction.

→ the variation of the excess carriers is LINEAR

→ @ contacts bulk concentration is kept

excess electrons in p-region $n_p' = n_p \exp\left(\frac{0.5V}{0.026}\right)$

$= 2.1\,10^5 \exp\left(\frac{0.5V}{0.026}\right)$

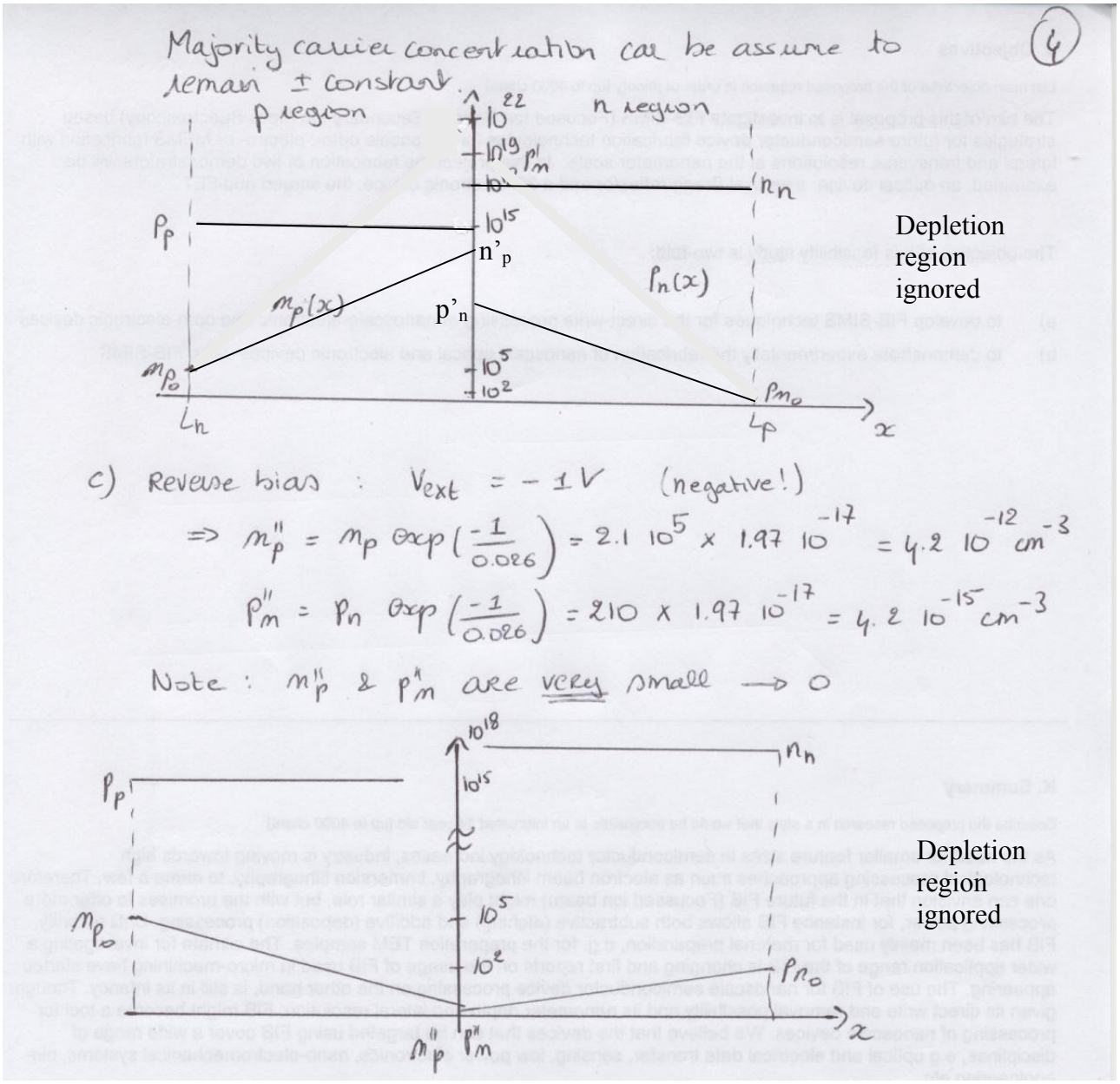
$\cong 4.7\,10^{13}\,cm^{-3}$

excess holes in n-region $p_n' = p_n \exp\left(\frac{0.5V}{0.026}\right)$

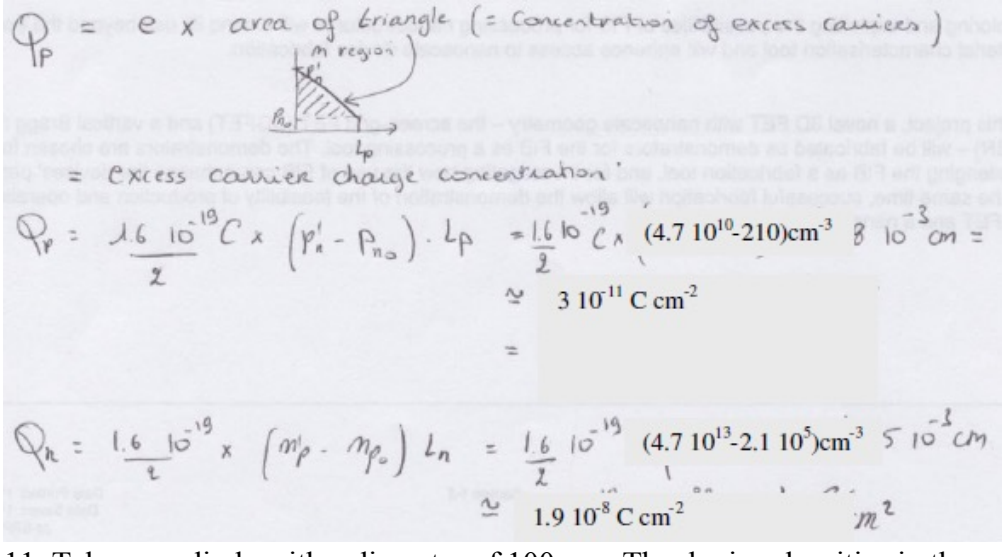
$= 210 \exp\left(\frac{0.5V}{0.026}\right)$

$\cong 4.7\,10^{10}\,cm^{-3}$

Note that the excess carrier concentration is a lot larger than the bulk minority carrier concentration.



10. Calculate the excess minority charge per area in each region of the diode of question 9.



11. Take a pn diode with a diameter of 100 μm. The doping densities in the p and n regions are respectively $N_A=10^{19}$ $cm^{-3}$and $N_D=10^{16}$ cm^{-3}. Calculate at T=300K the following:
    a. The built-in voltage V_0

b. The depletion width in the n and p doped material
c. Give a reason why the depletion width into the n-side is larger than into the p-side.
d. The charge Q in the depletion region at each side of the junction.
e. The minimum electric field.

$$w_p = \sqrt{\frac{2\varepsilon_0\varepsilon_r(V_0 - V)}{e}\left(\frac{N_D}{N_A(N_A + N_D)}\right)}$$

$$w_n = \sqrt{\frac{2\varepsilon_0\varepsilon_r(V_0 - V)}{e}\left(\frac{N_A}{N_D(N_A + N_D)}\right)}$$

$\varepsilon_0\varepsilon_r \frac{dE}{dx} = \rho(x)$ 1D Poisson equation ($\rho(x)$ is total charge concentration)

a) $V_0 = \frac{kT}{e} \ln\left(\frac{N_A N_D}{n_i^2}\right) = 0.026 \ln\left(\frac{10^{19}\ 10^{16}}{(1.45\ 10^{10})^2}\right) \simeq 0.88V$

b) $w_p = \sqrt{\frac{2\ 8.85\ 10^{-14}\ \frac{F}{cm}\ 11.9}{1.6\ 10^{-19}\ C}\left[\frac{10^{16}}{10^{19}\left[10^{19}+10^{16}\right]}\right] \times 0.88^{1}} \quad (V=0)$

$= \sqrt{1.16\ 10^{-15}}$

$= 3.4\ 10^{-8}\ cm$

Note $\frac{10^{16}}{10^{19}\left[10^{19}+10^{16}\right]} \simeq \frac{10^{16}}{10^{19} \times 10^{19}} = 10^{-22}\ cm^{-3}$

or $N_A >> N_D \Rightarrow \frac{N_D}{N_A(N_A+N_D)} \simeq \frac{N_D}{N_A^2}$

$w_n = \sqrt{\frac{2\ \varepsilon_0\ \varepsilon_R\ V_0}{e}\left[\frac{N_A}{N_D\left(N_A+N_D\right)}\right]} \simeq \sqrt{\frac{2\ \varepsilon_0 \varepsilon_r V_0}{e} \cdot \frac{1}{N_D}}$

($N_A >> N_D$)

$= \sqrt{\frac{2 \times 8.85\ 10^{-14}\ \frac{F}{cm} \times 11.9 \times 0.88V}{1.6\ 10^{-19}\ cm \times 10^{16}\ cm^{-3}}} = 34\ 10^{-6}\ cm$

$\Rightarrow$ The depletion width extends in the lowest doped Region!

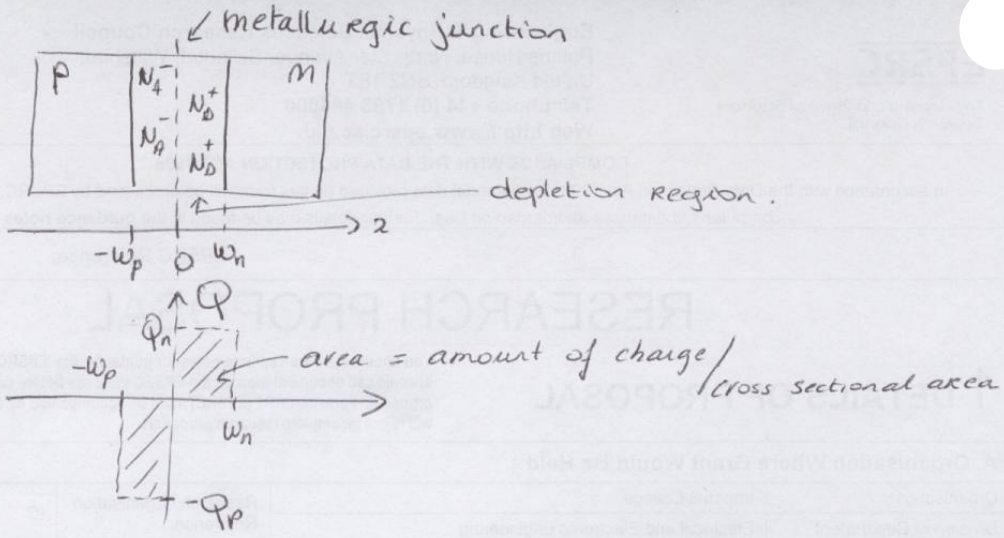
c) Charge neutrality across the depletion region requires that the charge per area due to the ionised acceptor atoms in the p-side depletion region = the charge per area due to the ionised donor atoms in the n-side depletion Region.

$\Rightarrow N_A \times w_p = N_D \times w_n$

when $N_A >> N_D$

$\Rightarrow w_p << w_n$

d)



$$\begin{cases} Q_n = e N_D w_n \cdot A \\ -Q_p = -e N_A w_p A \end{cases} \qquad A = \pi \left(\frac{d}{2}\right)^2 \qquad d = 10^{-2}\,cm$$

$$Q_n = 1.6\,10^{-19}\,C \times 10^{16}\,cm^{-3}\; 34\,10^{-6}\,cm\; \pi \left(\frac{10^{-2}}{2}\right)^2 cm^2$$

$$= 4.27\,10^{-12}\,C$$

$$Q_p = +Q_n = +4.27\,10^{-12}\,C \quad \text{charge neutrality.}$$

e) Write 1D Poisson equation in the n-type depletion region

$$\frac{d\mathcal{E}}{dx} = \frac{e N_D}{\varepsilon_0 \varepsilon_R} \quad \left(= \frac{\text{charges}}{\varepsilon} \text{ in } 0 < x < x_n\right)$$

($\frac{d\mathcal{E}}{dx}$: electric field)

$$\int_{\mathcal{E}_0}^{0} d\mathcal{E} = \int_0^{w_n} \frac{e N_D}{\varepsilon_0 \varepsilon_R}\, dx$$

$\mathcal{E}$-field = minimum @ $x = 0$

$\cong 0$ outside depletion region

$$\mathcal{E}_0 = -\frac{e N_D w_n}{\varepsilon_0 \varepsilon_R} = \frac{-1.6\,10^{-19}\,C \times 10^{16}\,cm^{-3} \times 34\,10^{-6}\,cm}{8.8\,10^{-14}\,\frac{F}{cm} \times 11.9}$$

$$= -51645\,\frac{V}{cm}$$

12. Calculate the hole and electron current for a bias of 0.5 V for the diode given in question 9. The length of the p and n region in the diode is the minority carrier diffusion length in these regions. Use the following parameters: D_n=35 cm^2/s, D_p=12.5 cm^2/s. Why is the electron current larger than the hole current? Give the total current through the diode.

For help with solving this equation quickly, look at the result of Q9b + using the approximation also used in Q.10.

Currents -> diffusion of minority carriers in pn diodes

$J_n = e\, D_n\, dn/dx \approx e\, D_n\, n'_p/L_n$
$J_p = e\, D_p\, dp/dx \approx e\, D_p\, p'_n/L_p$

Notes:

- Variation is linear, so gradient is constant and easy to calculate
- Excess minority carrier at depletion region edge (p'_n & n'_p) must larger value than equilibrium values (p_{n0} & n_{p0}), thus the latter are neglected in the gradient
- p'_n & n'_p are the result of Q9b (same voltage of 0.5V)

$J_n \approx 1.6\times10^{-19}$ C $35\times cm^2$ s^{-1} 4.7×10^{13} $cm^{-3}/(5\times10^{-3}$ cm) $\approx 5.3\times10^{-2}$ A cm^{-2}
$J_p \approx 1.6\times10^{-19}$ C $12.5\times cm^2$ s^{-1} 4.7×10^{10} $cm^{-3}/(8\times10^{-3}$ cm) $\approx 1.2\times10^{-5}$ A cm^{-2}

$J_p < J_n$ because $N_A < N_D$ making $n'_p > p'_n$

$J_{tot} \approx J_n$ Is an handy approximation for fast calculations

13. If the length of the n-region of the diode in question 12 is halved, what will happen to the total current?

Reducing the length of the material will increase the minority carrier *gradient* in that region. If the length is halved the gradient is doubled, thus the current flowing in the n-region increases with a factor of 2. Current in n-region is hole diffusion.
J_p new = J_p old × 2 = 2.4 10^{-5} A cm^{-2}
$J_{tot} \approx J_n$ no real change